

Enterprise Services Infrastructure Deployment

NETWORK ADMINISTRATION INTERNSHIP PROGRAM



NETWORKING

ITSimplera Institute

Week 05 Project Report

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CHAPTER 1

INTRODUCTION

1.1 Background

This report documents Week 5 of the Network Administration Internship Program at IT-Simplera Institute, undertaken under the supervision of Jawad Qayum, Senior Network Administrator. The internship provides hands-on exposure to enterprise network design, configuration, security hardening, and service deployment using Cisco Packet Tracer and GNS3 as simulation platforms.

Following the completion of Week 4, which established a segmented, access-controlled enterprise network across four VLANs (Human Resources, IT & Engineering, Finance, and a central Server Farm) on the gateway router R1-HQ-GW, Week 5 builds directly on that foundation. The objective of this assignment was to design, deploy, and verify a complete set of centralized enterprise services — DHCP, DNS, HTTP/HTTPS, FTP, TFTP, NTP, Syslog, and SNMP — that support automated network operations, secure communication, and efficient service delivery across the multi-department environment.

1.2 Objectives of the Assignment

- Design and deploy a centralized enterprise services infrastructure using Cisco Packet Tracer to support multiple departments within an organization.
- Configure and integrate essential enterprise network services — DHCP, DNS, HTTP/HTTPS, FTP, TFTP, NTP, Syslog, and SNMP — for reliable and efficient network operations.
- Implement secure access and service availability by applying appropriate configurations, access control policies, and enterprise networking best practices.
- Verify the functionality and interoperability of all configured services through comprehensive testing, monitoring, and end-to-end connectivity validation.
- Troubleshoot and resolve service-related issues to maintain continuous network operation and demonstrate effective enterprise service management skills.

1.3 Scope of the Report

This report is organized into seven chapters. Chapter 1 introduces the assignment and the core protocols involved. Chapter 2 profiles IT-Simplera Institute and the Network Administration department. Chapter 3 details the network topology and addressing scheme carried forward from Week 4. Chapter 4 documents the configuration of each of the eight enterprise services. Chapter 5 presents verification, connectivity testing, and troubleshooting. Chapter 6 summarizes the tools, technologies, and skills applied, and Chapter 7 concludes with challenges faced, lessons learned.

1.4 Overview of Enterprise Network Protocols & Services

The following protocols form the technical backbone of this assignment. Each is defined below with its transport-layer characteristics and role within the enterprise services infrastructure, before being configured and verified in detail in Chapter 4.

Protocol	Full Name	Transport / Port	Primary Function
DHCP	Dynamic Host Configuration Protocol	UDP 67 (server) / 68 (client)	Automatic IP address, subnet mask, gateway and DNS assignment to hosts
DNS	Domain Name System	UDP/TCP 53	Resolves human-readable hostnames to IP addresses
HTTP	Hypertext Transfer Protocol	TCP 80	Delivers unencrypted web content between client and server
HTTPS	HTTP Secure	TCP 443	Delivers web content encrypted with SSL/TLS
FTP	File Transfer Protocol	TCP 21 (control) / 20 (data)	Authenticated upload/download of files between hosts
TFTP	Trivial File Transfer Protocol	UDP 69	Lightweight, connectionless file transfer — primarily IOS image and configuration backup/restore
NTP	Network Time Protocol	UDP 123	Synchronizes device clocks across the network to a common time source
Syslog	System Logging Protocol	UDP 514	Centralizes event, error and audit log messages from network devices
SNMP	Simple Network Management Protocol	UDP 161 (poll) / 162 (trap)	Enables remote monitoring and management of network devices via MIB objects

DHCP — Dynamic Host Configuration Protocol

DHCP eliminates manual IP configuration by leasing addresses to clients from a defined scope. It operates using the four-step DORA process — Discover, Offer, Request, and Acknowledge — and also distributes default gateway, DNS server, and lease-time information. In a routed, multi-VLAN network, a centralized DHCP server requires an ip helper-address (DHCP relay) configured on each gateway sub-interface so that client broadcasts can reach the server across subnet boundaries.

DNS — Domain Name System

DNS provides hierarchical name resolution, allowing users to reach services using memorable hostnames (e.g. `www.hq.local`) instead of raw IP addresses. A DNS server hosts resource records — primarily A records mapping a name to an IPv4 address — within a configured domain zone, and responds to client queries issued on UDP (or TCP for larger responses) port 53.

HTTP / HTTPS — Web Services

HTTP is the application-layer protocol used to serve intranet web pages to client browsers on TCP port 80. HTTPS wraps the same HTTP exchange inside a TLS/SSL encrypted tunnel on TCP port 443, protecting page content and credentials from interception. Enterprise intranet portals are typically configured to support both, with HTTPS preferred for any authenticated or sensitive content.

FTP — File Transfer Protocol

FTP provides authenticated, bidirectional file transfer between a client and server using a separate control connection (TCP 21) and data connection (TCP 20). User accounts are bound to specific permissions — read, write, delete, list — which makes FTP suitable for managed file-sharing and configuration-backup repositories within an enterprise.

TFTP — Trivial File Transfer Protocol

TFTP is a simplified, connectionless file-transfer protocol running over UDP port 69 with no authentication. Its simplicity makes it the standard mechanism for transferring Cisco IOS images and for backing up / restoring router and switch running-configurations (`copy running-config tftp`).

NTP — Network Time Protocol

NTP synchronizes the system clocks of routers, switches, and servers to a common reference time source, arranged in a stratum hierarchy. Accurate, synchronized timestamps are essential for correlating Syslog events, SNMP traps, and security audit trails across multiple devices.

Syslog — Centralized Logging

Syslog is the standard for forwarding event, warning, and error messages generated by network devices to a centralized logging host over UDP port 514. Centralized logging allows administrators to audit configuration changes, detect anomalies, and troubleshoot incidents from a single console rather than each device individually.

SNMP — Simple Network Management Protocol

SNMP allows a network management station to poll devices for operational statistics (interface counters, CPU/memory load, uptime) via GET requests on UDP port 161, and allows devices to push unsolicited traps to the manager on UDP port 162. Access is controlled using community strings (SNMPv1/v2c) or user-based authentication and encryption (SNMPv3).

CHAPTER 2

ORGANIZATION PROFILE

2.1 About the Organization

IT-Simplera Institute is an IT training and solutions organization that runs a structured Network Administration Internship Program, providing interns with practical, lab-based exposure to enterprise networking concepts through weekly, progressively advanced assignments delivered in Cisco Packet Tracer and GNS3.

2.2 Department / Team

The internship is carried out within the Network Administration department, under the direct supervision of **Jawad Qayum**, Senior Network Administrator. The department is responsible for setting weekly technical assignments, reviewing submitted documentation and lab files, and conducting evaluation interviews on Google Meet to assess each intern's understanding, implementation quality, and problem-solving ability.

2.3 Organizational Network Infrastructure

The lab environment simulated during this internship models a single-site enterprise network. In Week 4, the network was segmented into four VLANs — Human Resources, IT & Engineering, Finance, and a central Server Farm — interconnected via 802.1Q sub-interfaces on a single gateway router (R1-HQ-GW), with Extended ACLs enforcing departmental isolation and management-plane lockdown. Week 5 extends this same topology by populating the Server Farm VLAN with the full suite of enterprise services described in this report.

CHAPTER 3

NETWORK TOPOLOGY & SERVICE DESIGN

3.1 Project Overview

The Week 5 deployment builds directly on the four-VLAN architecture and Extended ACL hardening completed in Week 5. Rather than starting from a blank topology, the existing Server Farm VLAN (VLAN 100) was populated with dedicated servers hosting DHCP, DNS, HTTP/HTTPS, FTP/TFTP, and the combined NTP/Syslog/SNMP management services, so that every department can consume centralized services through the same gateway router, R1-HQ-GW.

3.2 Network Topology

The diagram below shows R1-HQ-GW, the access-layer switches for each VLAN, the Server Farm, and all end devices used to validate the deployment.

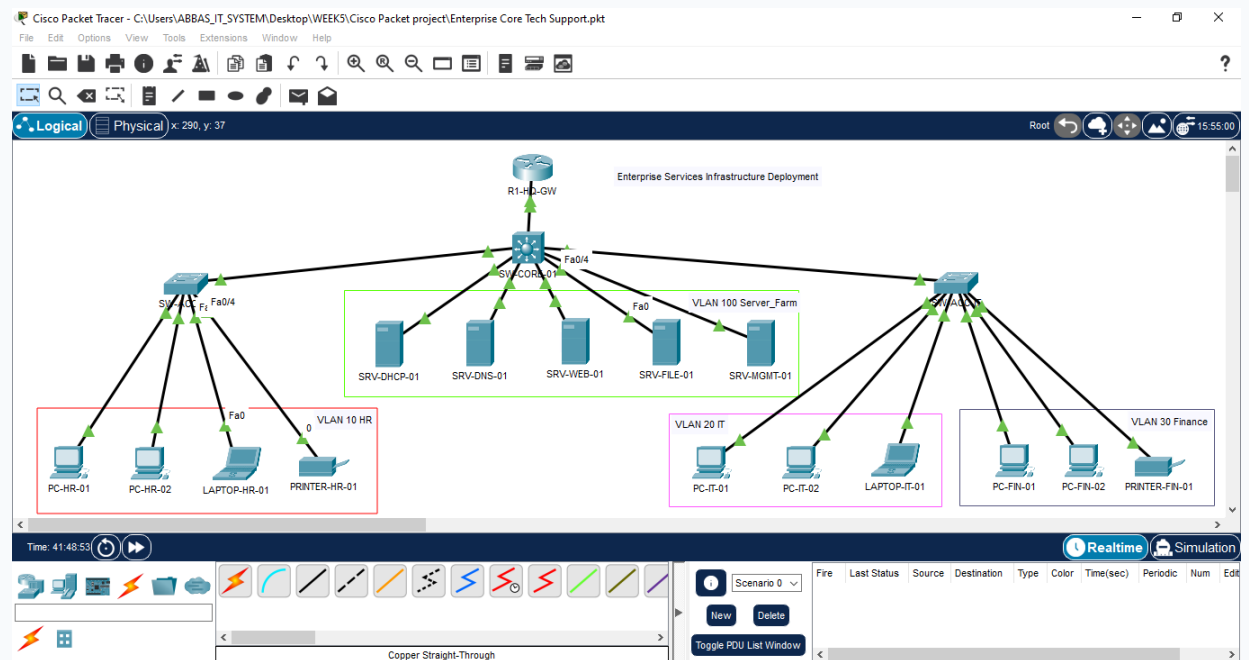


Figure 1: Complete enterprise topology in Cisco Packet Tracer — R1-HQ-GW, VLAN switches, Server Farm, and end devices

3.3 VLAN & IP Addressing Scheme

The department-facing VLANs and addressing plan are unchanged from Week 4. The table below summarizes the scheme, with the Server Farm row expanded to show every service host added in Week 5.

VLAN	Department / Purpose	Subnet	Gateway (Sub-Int)
10	Human Resources (HR)	10.10.10.0/24	Gi0/0/0.10 — 10.10.10.1
20	IT & Engineering	10.10.20.0/24	Gi0/0/0.20 — 10.10.20.1
30	Finance & Accounts	10.10.30.0/24	Gi0/0/0.30 — 10.10.30.1
100	Server Farm & Management	10.10.100.0/24	Gi0/0/0.100 — 10.10.100.1

Server Farm host assignments used for the Week 5 service deployment:

Server	IP Address	Service(s) Hosted	Access Scope
SRV-DHCP	10.10.100.10	DHCP	All departments (HR, IT, Finance)
SRV-DNS	10.10.100.20	DNS (zone: itsimplera.local)	All departments
SRV-WEB	10.10.100.30	HTTP / HTTPS	All departments
SRV-FILE	10.10.100.40	FTP / TFTP	All departments
SRV-MGMT	10.10.100.50	NTP / Syslog / SNMP	IT (VLAN 20) only — RESTRICT_HR / RESTRICT_FINANCE deny

3.4 Security Foundation Week 5

Two Extended ACLs, RESTRICT_HR and RESTRICT_FINANCE, remain bound inbound on sub-interfaces Gi0/0/0.10 and Gi0/0/0.30 respectively. They continue to block direct HR ↔ Finance traffic and deny both departments access to the management host 10.10.100.50, while permitting all other traffic — including the new DHCP, DNS, Web, and File services. This means the Week 5 service rollout inherits the zero-trust departmental isolation already validated in Week 4 without any additional rule changes, other than confirming NTP, Syslog and SNMP (all hosted on 10.10.100.50) correctly remain restricted to the IT department.

CHAPTER 4

SERVICE IMPLEMENTATION & CONFIGURATION

Each of the eight enterprise services was configured on its dedicated Server Farm host and, where relevant, on R1-HQ-GW. Configuration steps, the router-side IOS commands used, and the verification approach are documented for every service below. Insert the corresponding screenshot from your own Packet Tracer lab in each placeholder.

4.1 DHCP Service Configuration

SRV-DHCP (10.10.100.20) was configured with three address pools, one per department VLAN. Because the server sits in a different subnet (VLAN 100) from the clients it serves, an ip-helper-address was added to each department's gateway sub-interface on R1-HQ-GW so that DHCP broadcast requests are relayed as unicast to the server.

```
! R1-HQ-GW – enable DHCP relay on each department sub-interface
interface GigabitEthernet0/0/0.10
 ip helper-address 10.10.100.20
exit
interface GigabitEthernet0/0/0.20
 ip helper-address 10.10.100.20
exit
interface GigabitEthernet0/0/0.30
 ip helper-address 10.10.100.20
exit
```

Server-side DHCP pool parameters configured on SRV-DHCP:

Pool Name	Network / Mask	Default Gateway	DNS Server	Start Address	Excluded / Static
POOL_HR	10.10.10.0 / 255.255.255.0	10.10.10.1	10.10.100.20	10.10.10.100	.1–.99 excluded
POOL_IT	10.10.20.0 / 255.255.255.0	10.10.20.1	10.10.100.20	10.10.20.100	.1–.99 excluded
POOL_FINANCE	10.10.30.0 / 255.255.255.0	10.10.30.1	10.10.100.20	10.10.30.100	.1–.99 excluded

SRV-DHCP-01

Physical Config **Services** Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface

FastEthernet0

Service ☒ On ☐ Off

Pool Name

serverPool

Default Gateway

10.10.100.1

DNS Server

10.10.100.20

Start IP Address :

10

10

100

0

Subnet Mask:

255

255

255

0

Maximum Number of Users :

256

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	10.10.10...	10.10.10...	10.10.10...	255.255....	256	0.0.0.0	0.0.0.0
POOL_IT	10.10.20.1	10.10.10...	10.10.20...	255.255....	156	0.0.0.0	0.0.0.0
POOL_FINANCE	10.10.30.1	10.10.10...	10.10.30...	255.255....	156	0.0.0.0	0.0.0.0
POOL_HR	10.10.10.1	10.10.10...	10.10.10...	255.255....	156	0.0.0.0	0.0.0.0

☐ Top

Figure 2: SRV-DHCP pool configuration screen (POOL_HR / POOL_IT / POOL_FINANCE)

9

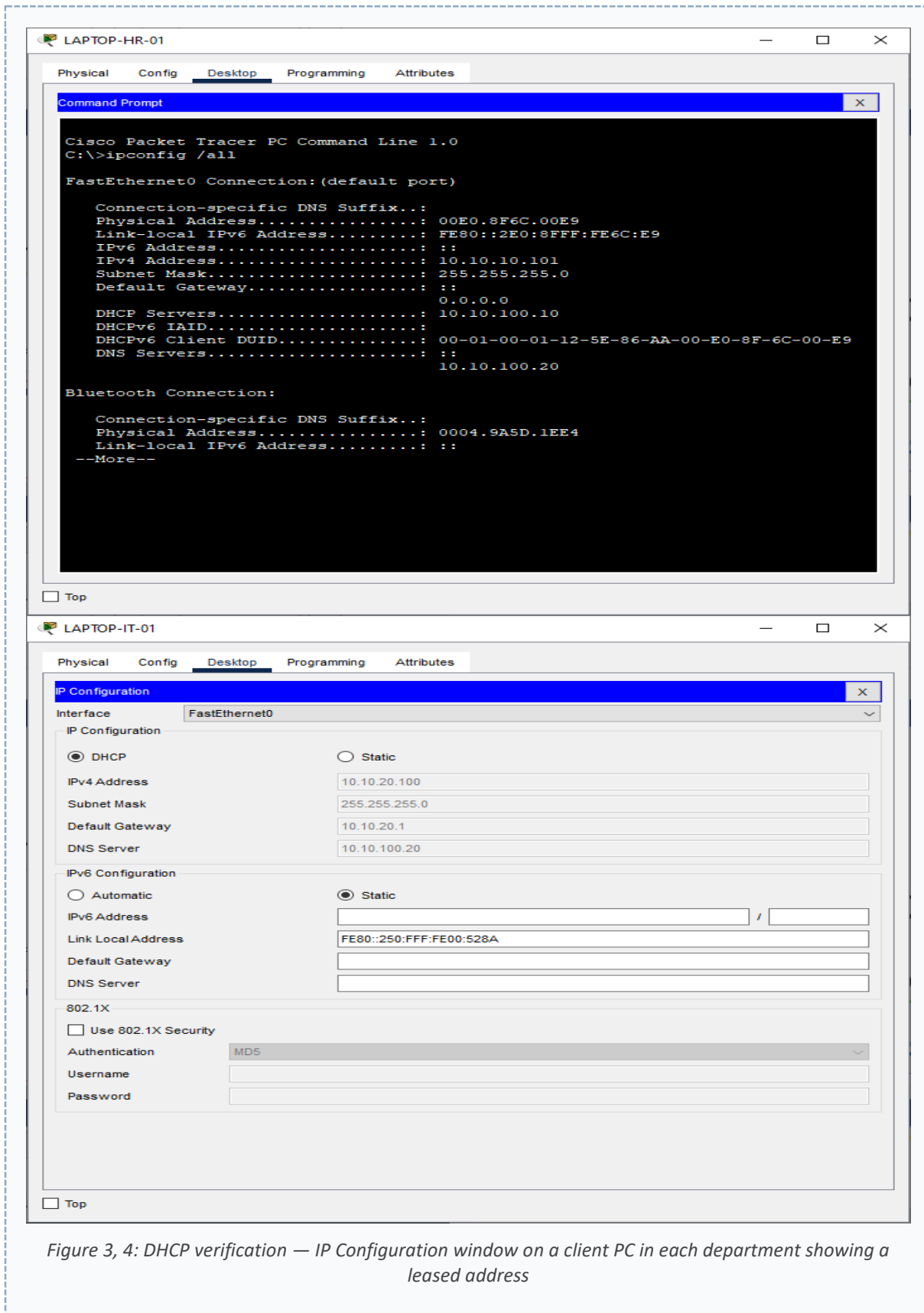


Figure 3, 4: DHCP verification — IP Configuration window on a client PC in each department showing a leased address

Verification: on each client, run `ipconfig /all` (or open the Desktop → IP Configuration tab) and confirm an address from the correct pool, along with the expected gateway and DNS server, has been assigned automatically.

4.2 DNS Service Configuration

SRV-DNS (10.10.100.20) hosts the `itsimplera.local` zone with A records resolving the hostname of every centralized service to its IP address, allowing users and administrators to reach services by name rather than by raw IP address.

Record Type	Hostname	Resolves To
A	itsimplera.local	10.10.100.30
CNAME	www.itsimplera.local	Itsimplera.local

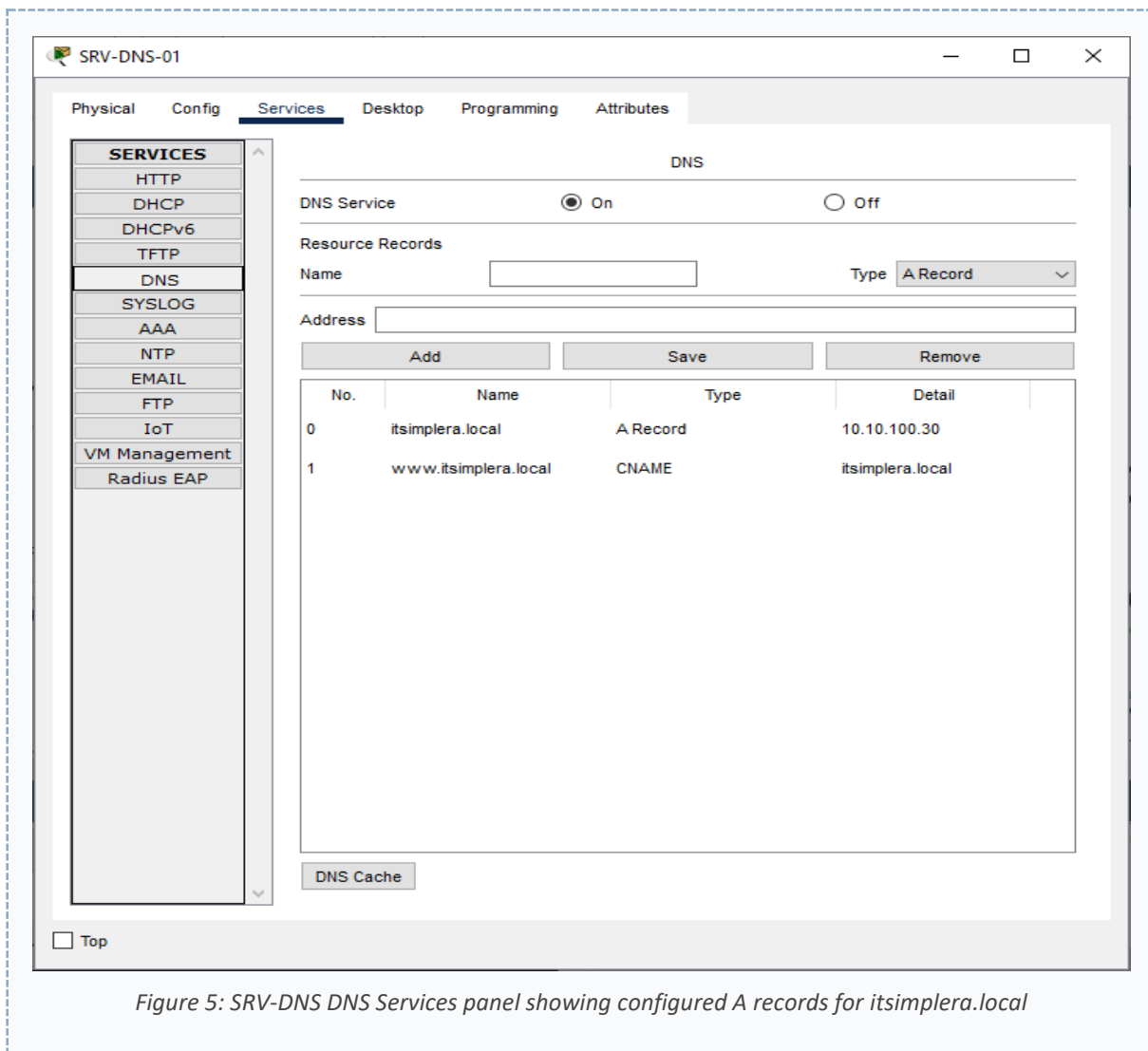


Figure 5: SRV-DNS DNS Services panel showing configured A records for `itsimplera.local`

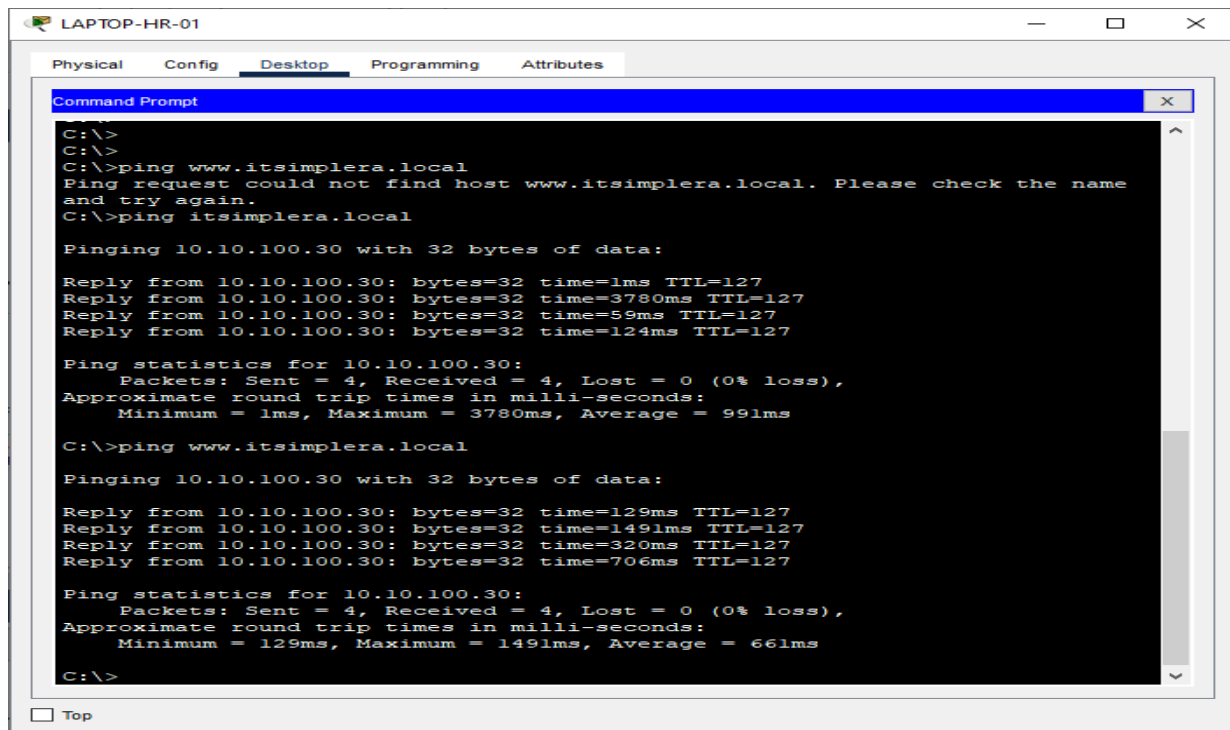
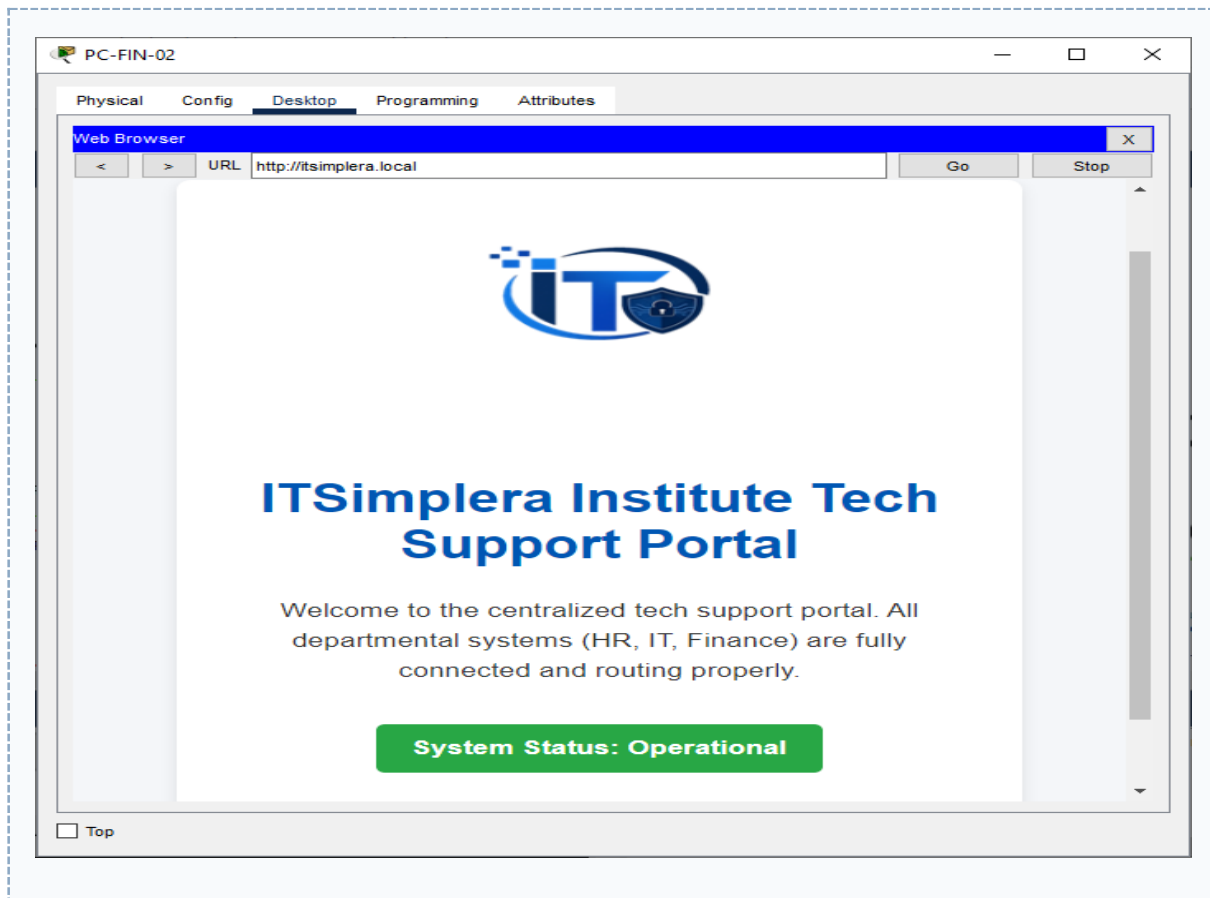


Figure 6,7: DNS verification — client Command Prompt resolving and pinging www.itsimplera.local by name

Verification: from a client PC, ping www.itsimplera.local and confirm the ping resolves and replies from 10.10.100.30, demonstrating successful hostname resolution.

4.3 HTTP / HTTPS Web Service Configuration

SRV-WEB (10.10.100.30) was configured with both HTTP and HTTPS services enabled in the server's Services → HTTP tab, serving a simple intranet homepage (index.html). Because this is a shared informational resource, it remains reachable from HR, IT, and Finance under the existing permit ip ... any rules in RESTRICT_HR / RESTRICT_FINANCE.

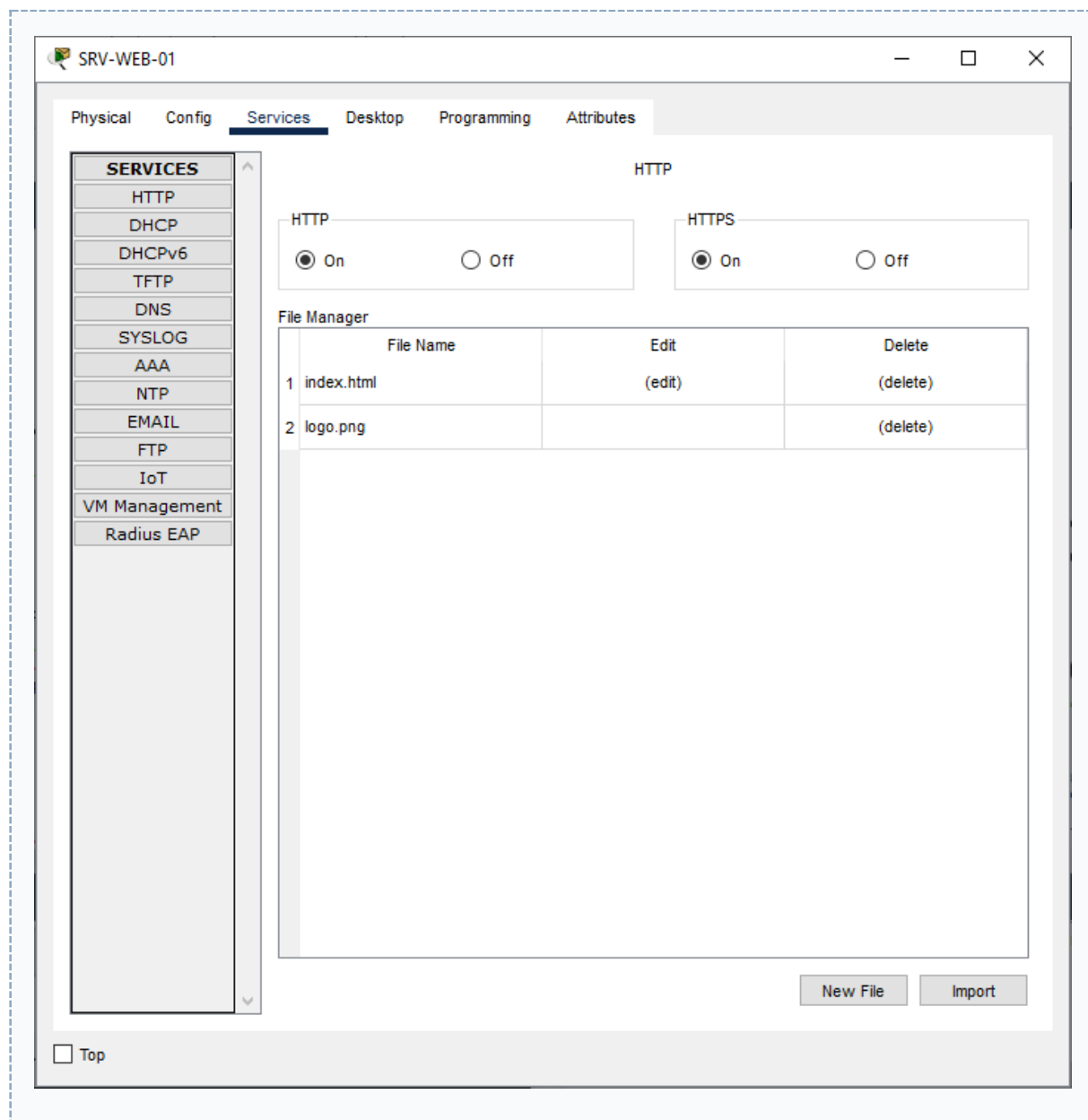


Figure 8: SRV-WEB HTTP/HTTPS service configuration panel (both toggled On)

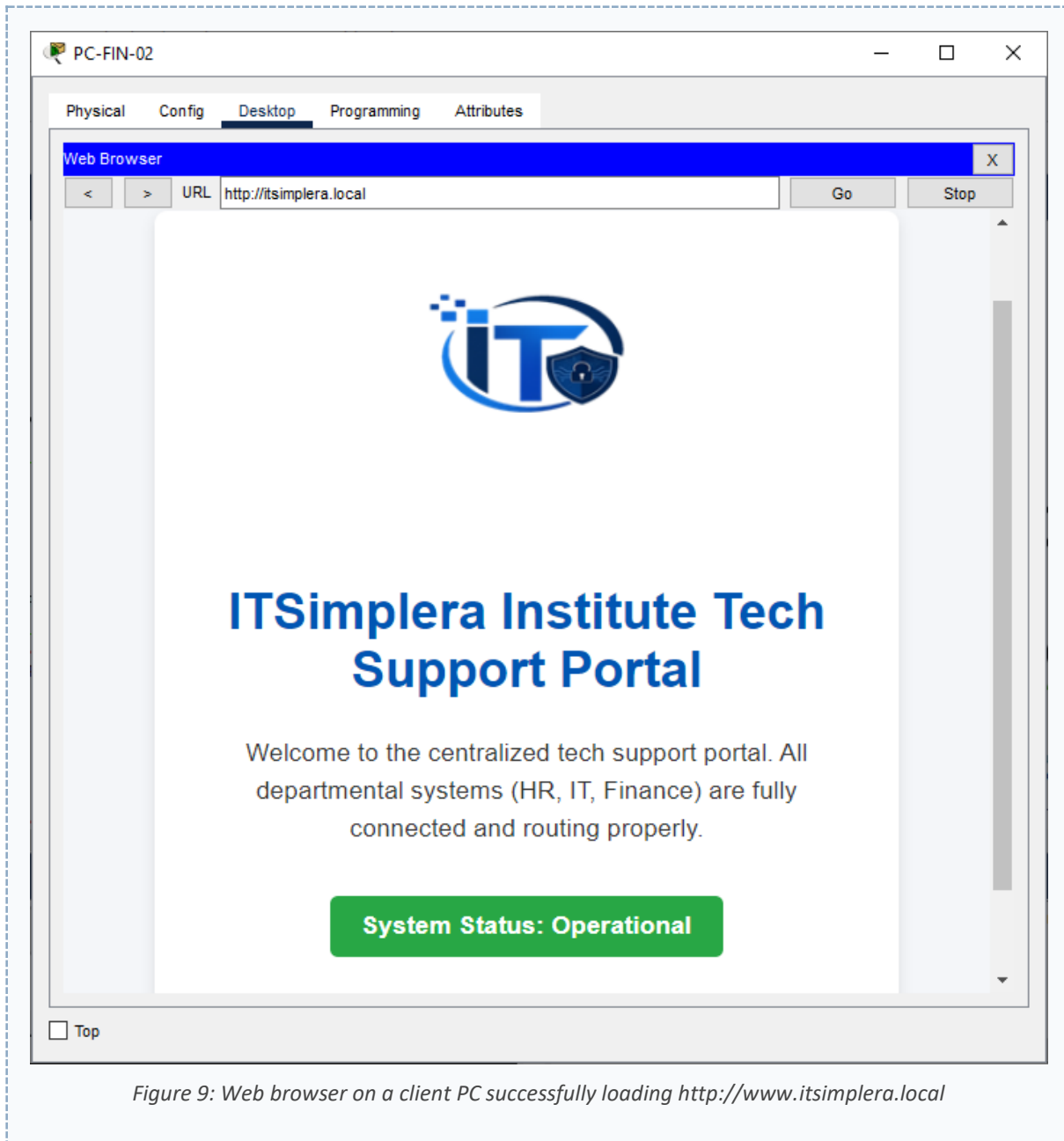


Figure 9: Web browser on a client PC successfully loading <http://www.itsimplera.local>

Verification: from a browser on any department client, load <http://www.itsimplera.local> or the direct IP, 10.10.100.30 and confirm the intranet page renders over protocol.

4.4 FTP Service Configuration

The FTP service on SRV-FILE (10.10.100.40) was enabled with a dedicated service account used for configuration-backup uploads and downloads.

Username	Password	Permissions
admin	Cisco123	Read / Write / Delete / Rename / List

The screenshot shows the SRV-FILE-01 configuration window with the 'Services' tab selected. The 'FTP' service is enabled (On). Under 'User Setup', the 'admin' user is configured with password 'cisco123' and permissions for Write, Read, Delete, Rename, and List. A table lists the configured users and their permissions.

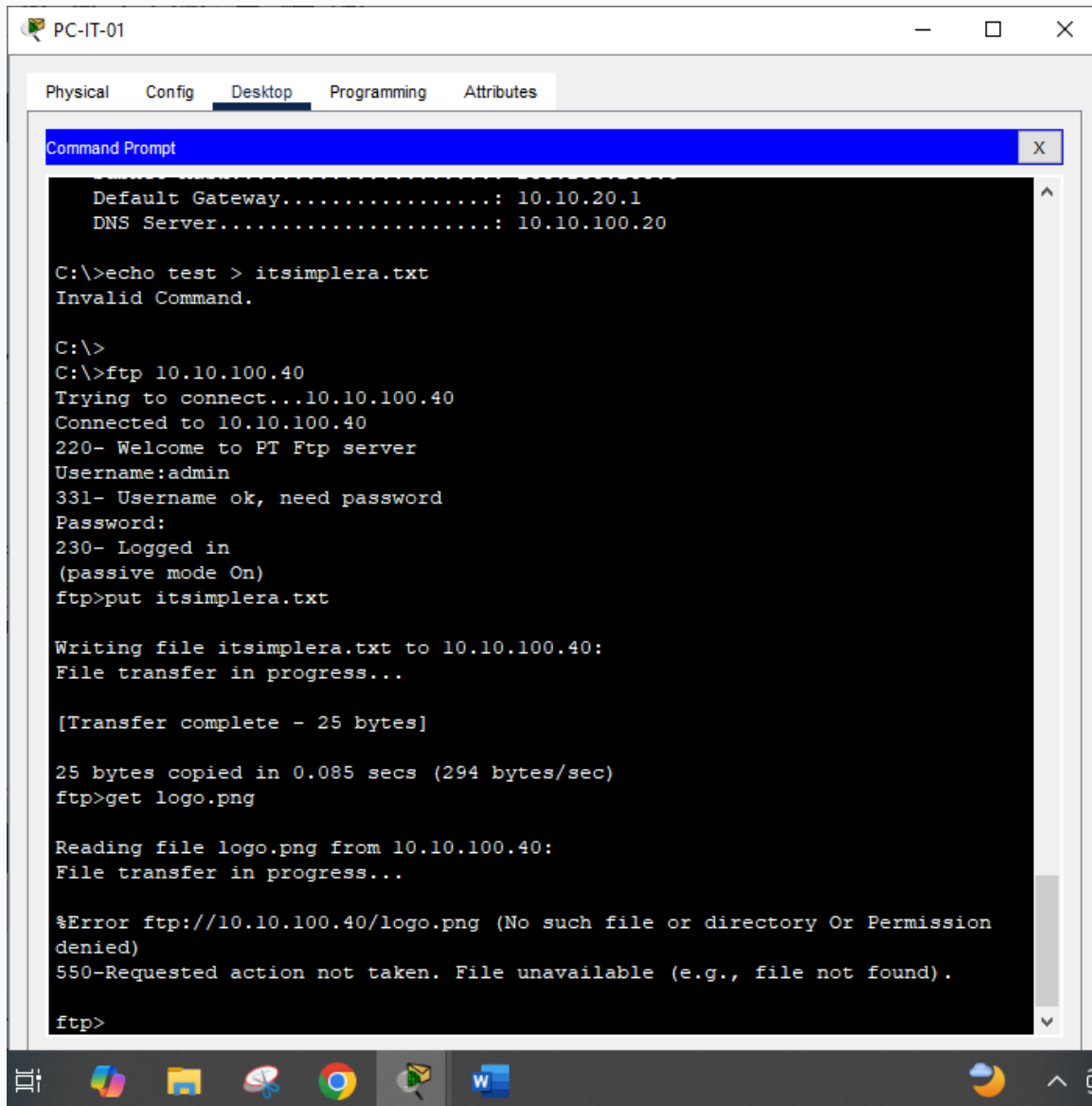
	Username	Password	Permission
1	admin	cisco123	RWDNL
2	cisco	cisco	RWDNL

Below the user setup, a 'File' section lists several files for removal:

- 1 asa842-k8.bin
- 2 asa923-k8.bin
- 3 c1841-advipservicesk9-mz.124-15.T1.bin
- 4 c1841-ipbase-mz.123-14.T7.bin
- 5 c1841-ipbasek9-mz.124-12.bin
- 6 c1900-universalk9-mz.SPA.155-3.M4a.bin

Buttons for 'Add', 'Save', 'Remove', and 'Top' are visible.

Figure 10: SRV-FILE FTP user account and permissions configuration



```
PC-IT-01
Physical  Config  Desktop  Programming  Attributes

Command Prompt

Default Gateway.....: 10.10.20.1
DNS Server.....: 10.10.100.20

C:\>echo test > itsimplera.txt
Invalid Command.

C:\>
C:\>ftp 10.10.100.40
Trying to connect...10.10.100.40
Connected to 10.10.100.40
220- Welcome to PT Ftp server
Username:admin
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>put itsimplera.txt

Writing file itsimplera.txt to 10.10.100.40:
File transfer in progress...

[Transfer complete - 25 bytes]

25 bytes copied in 0.085 secs (294 bytes/sec)
ftp>get logo.png

Reading file logo.png from 10.10.100.40:
File transfer in progress...

%Error ftp://10.10.100.40/logo.png (No such file or directory Or Permission
denied)
550-Requested action not taken. File unavailable (e.g., file not found).

ftp>
```

Figure 11: FTP client session from PC-IT-01 — successful login, file upload (put) and download (get)

Verification: from PC-IT-01, open a Command Prompt and connect with ftp 10.10.100.40, log in with the admin credentials, and confirm put and get operations complete successfully against the server's file directory.

4.5 TFTP Service Configuration

TFTP was enabled alongside FTP on SRV-FILE and used specifically for router and switch configuration backup and restore, reflecting its real-world role in IOS image and configuration management.

```
! R1-HQ-GW – backup running-config to the TFTP server
```



```
R1-HQ-GW# copy running-config tftp
Address or name of remote host []? 10.10.100.40
Destination filename [r1-hq-gw-config]?
!!
1978 bytes copied in 0.512 secs
```

```
! Restore, if required
R1-HQ-GW# copy tftp running-config
Address or name of remote host []? 10.10.100.40
Source filename []? r1-hq-gw-backup.cfg
```

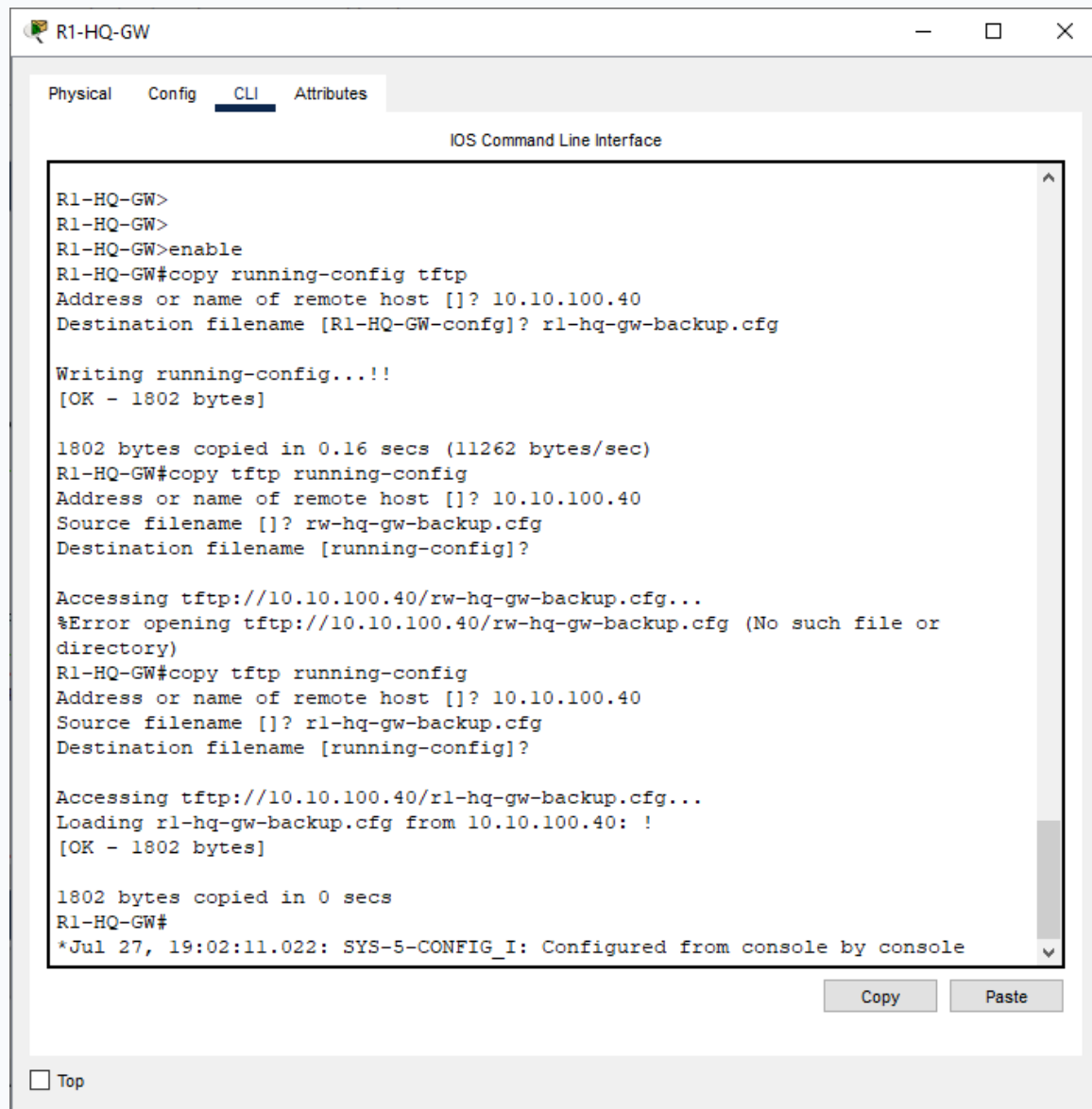


Figure 12: TFTP backup/restore — CLI output of copy running-config tftp against SRV-FILE

Verification: confirm the backup file appears in the SRV-FILE TFTP root directory after the copy running-config tftp operation, and that a restore (copy tftp running-config) successfully re-applies it.

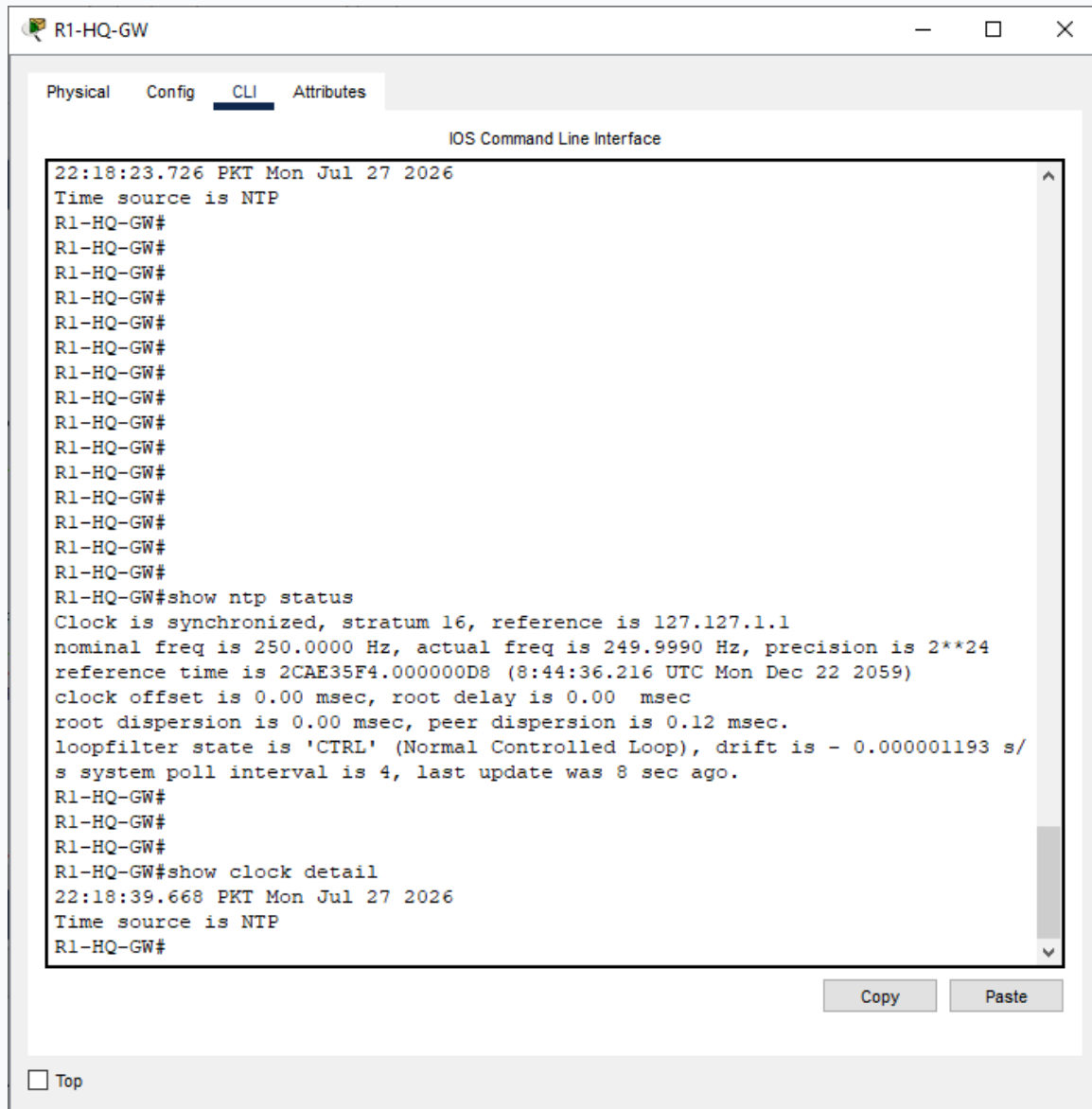
4.6 NTP Service Configuration

SRV-MGMT (10.10.100.50) was configured as the authoritative NTP time source for the network. R1-HQ-GW was configured as an NTP client pointing to this server, keeping all logged events across the network on a consistent timeline.

```
! R1-HQ-GW – NTP client configuration
ntp server 10.10.100.50
clock timezone PKT 3 0
```

The screenshot shows the configuration interface for SRV-MGMT-01, specifically the **Services** tab. The **NTP** service is selected in the left-hand menu. The main configuration area shows the NTP service is **On**. The **Authentication** section has **Disable** selected for the **Enable** radio button. The **Key** and **Password** fields are empty. Below this is a calendar for **July, 2026**, showing the days of the week and the dates. The current time is **07:02:54PM**. At the bottom left, there is a **Top** button.

Sun	Mon	Tue	Wed	Thu	Fri	Sat
28	29	30	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1
2	3	4	5	6	7	8



The screenshot shows a terminal window titled "R1-HQ-GW" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The output shows the time source is NTP, followed by a series of "R1-HQ-GW#" prompts. The command "show ntp status" is entered, resulting in the following output: "Clock is synchronized, stratum 16, reference is 127.127.1.1 nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24 reference time is 2CAE35F4.000000D8 (8:44:36.216 UTC Mon Dec 22 2059) clock offset is 0.00 msec, root delay is 0.00 msec root dispersion is 0.00 msec, peer dispersion is 0.12 msec. loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system poll interval is 4, last update was 8 sec ago." This is followed by more "R1-HQ-GW#" prompts and the command "show clock detail", which outputs: "22:18:39.668 PKT Mon Jul 27 2026 Time source is NTP R1-HQ-GW#". At the bottom of the terminal window, there are "Copy" and "Paste" buttons and a "Top" button.

```
22:18:23.726 PKT Mon Jul 27 2026
Time source is NTP
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#show ntp status
Clock is synchronized, stratum 16, reference is 127.127.1.1
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is 2CAE35F4.000000D8 (8:44:36.216 UTC Mon Dec 22 2059)
clock offset is 0.00 msec, root delay is 0.00 msec
root dispersion is 0.00 msec, peer dispersion is 0.12 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/
s system poll interval is 4, last update was 8 sec ago.
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#show clock detail
22:18:39.668 PKT Mon Jul 27 2026
Time source is NTP
R1-HQ-GW#
```

Figure 13,14: SRV-MGMT NTP service panel (enabled) and R1-HQ-GW show ntp status output confirming synchronization

Because NTP traffic to 10.10.100.50 is subject to the same management-lockdown ACLs as Syslog and SNMP, only the IT VLAN can query SRV-MGMT directly; HR and Finance hosts continue to rely on R1-HQ-GW as their synchronized time reference, which is expected behaviour consistent with the Week 5 security design.

4.7 Syslog Service Configuration

R1-HQ-GW was configured to forward all log messages of severity 6 (informational) and above to the centralized Syslog service running on SRV-MGMT, allowing configuration changes and interface events to be audited from a single console.

```
! R1-HQ-GW – centralized logging configuration
logging host 10.10.100.50
logging trap informational
logging on
service timestamps log datetime msec
```

The screenshot shows the configuration interface for SRV-MGMT-01. The 'Services' tab is selected, and the 'Syslog' service is configured. The 'Syslog' service is turned 'On'. The 'Service' list on the left includes HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG (selected), AAA, NTP, EMAIL, FTP, IoT, VM Management, and Radius EAP. The main area displays a table of log entries.

Service	Time	HostName	Message
1	07.27.2026 10:25:28.856 AM	10.10.100.1	%SYS-5-CONFIG_I: Configured from console b...
2	07.27.2026 10:25:28.856 AM	10.10.100.1	: %SYS-6-LOGGINGHOST_STARTST...
3	07.27.2026 10:37:38.440 AM	10.10.100.1	%SYS-5-CONFIG_I: Configured from console b...
4	07.27.2026 11:49:04.480 AM	10.10.100.1	%SYS-5-CONFIG_I: Configured from console b...
5	07.27.2026 05:43:35.113 PM	10.10.100.1	%SYS-5-CONFIG_I: Configured from console b...
6	07.27.2026 06:13:44.704 PM	10.10.100.1	%SYS-5-CONFIG_I: Configured from console b...
7	07.27.2026 06:15:13.392 PM	10.10.100.1	%SYS-5-CONFIG_I: Configured from console b...
8	07.27.2026 07:02:11.254 PM	10.10.100.1	%SYS-5-CONFIG_I: Configured from console b...
9	07.27.2026 07:05:41.600 PM	10.10.100.1	%SYS-5-CONFIG_I: Configured from console b...
10	07.27.2026 07:10:32.001 PM	10.10.100.1	%SYS-5-CONFIG_I: Configured from console b...
11	07.27.2026 07:11:20.564 PM	10.10.100.1	%SYS-5-CONFIG_I: Configured from console b...
12	07.27.2026 07:14:26.524 PM	10.10.100.1	%SYS-5-CONFIG_I: Configured from console b...
13	07.27.2026 07:17:56.083 PM	10.10.100.1	%SYS-5-CONFIG_I: Configured from console b...

Clear Log

Top

Figure 15: SRV-MGMT Syslog panel showing received log entries, e.g. %SYS-5-CONFIG_I: Configured from console

Verification: make a small configuration change on R1-HQ-GW (e.g., enter and exit configuration mode) and confirm the corresponding %SYS-5-CONFIG_I event appears on the SRV-MGMT Syslog panel with an accurate, NTP-synchronized timestamp.

4.8 SNMP Service Configuration

SNMP was enabled on R1-HQ-GW with a read-only community string, and SRV-MGMT was configured as both the polling manager and trap destination, enabling remote monitoring of interface status, CPU load, and uptime.

```
! R1-HQ-GW - SNMP configuration
snmp-server community public RO
snmp-server community private RW
snmp-server host 10.10.100.50 public
snmp-server enable traps
```

```
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#
R1-HQ-GW#show running-config | include snmp
snmp-server community public RO
snmp-server community private RW
R1-HQ-GW#
```

Copy

Paste

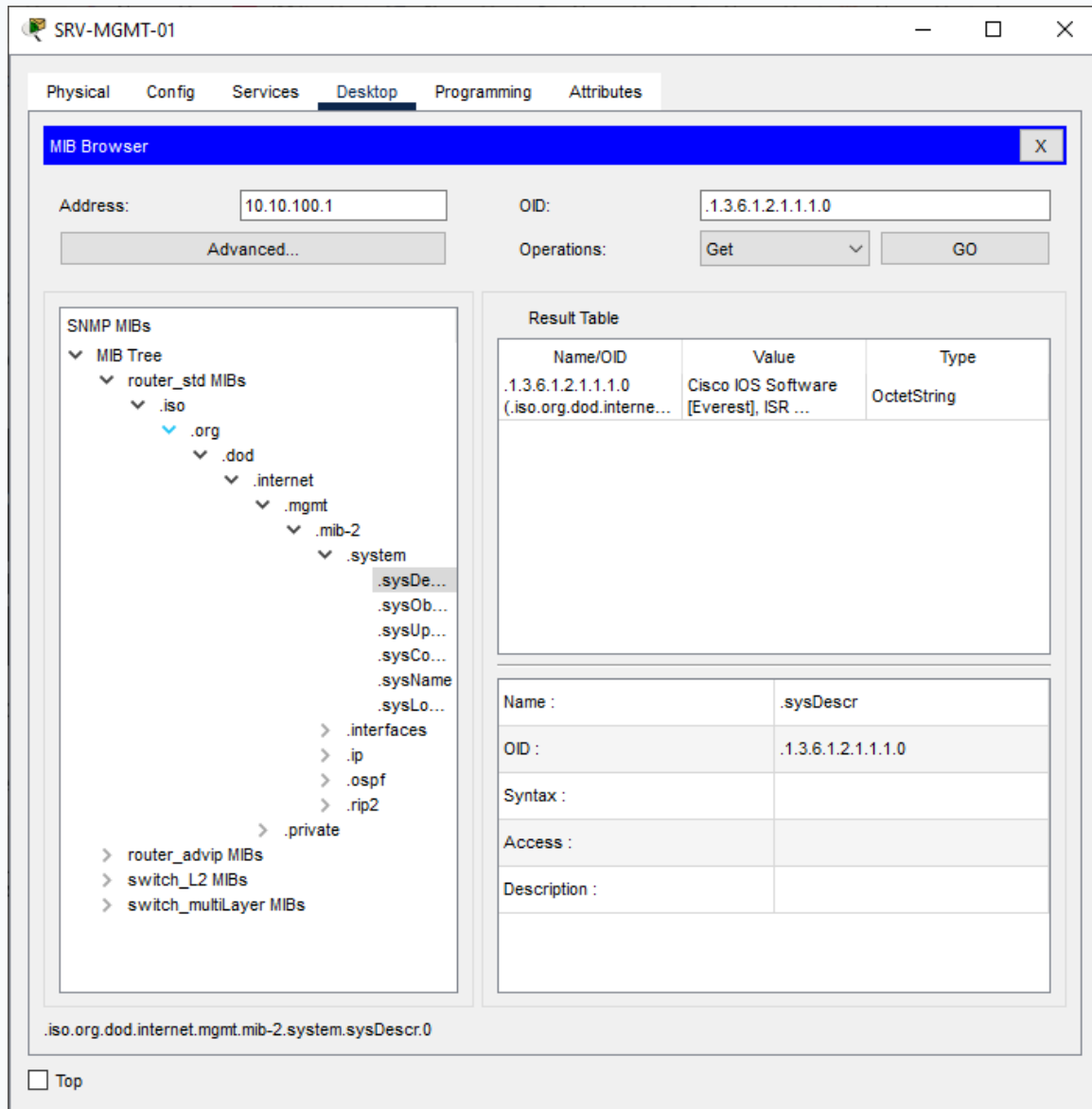


Figure 16,17: SRV-MGMT SNMP / MIB browser panel successfully polling R1-HQ-GW device statistics

Verification: from the SRV-MGMT SNMP management panel, poll R1-HQ-GW using the public community string and confirm interface counters and system uptime are returned successfully.

CHAPTER 5

VERIFICATION, TESTING & TROUBLESHOOTING

5.1 Connectivity & Service Verification Matrix

End-to-end tests were performed from a representative host in each department against every centralized service, consistent with the Week 5 ACL design (HR and Finance denied from 10.10.100.50; all departments permitted to DHCP, DNS, Web, and File services).

Source	Destination / Service	Test	Expected Result
PC-HR-01	SRV-DHCP 10.10.100.20	DHCP lease	Address assigned from HR_POOL
PC-HR-01	SRV-DNS 10.10.100.20	nslookup / ping www.itsimplera.local	Resolves to 10.10.100.30
PC-HR-01	SRV-WEB 10.10.100.30	HTTP & HTTPS	Intranet page loads on both
PC-HR-01	SRV-FILE 10.10.100.40	FTP login	Permitted — shared resource
PC-HR-01	SRV-MGMT 10.10.100.50	Ping / NTP / Syslog / SNMP	Blocked — RESTRICT_HR (ACL)
PC-FIN-01	SRV-MGMT 10.10.100.50	Ping / NTP / Syslog / SNMP	Blocked — RESTRICT_FINANCE (ACL)
PC-FIN-01	SRV-DHCP / DNS / WEB / FILE	Full-service access	Permitted
PC-IT-01	SRV-MGMT 10.10.100.50	Ping / NTP / Syslog / SNMP	Permitted — admin access
PC-IT-01	All Server Farm hosts	Full-service access	Permitted
PC-HR-01	PC-FIN-01	Inter-VLAN ping	Blocked — RESTRICT_HR

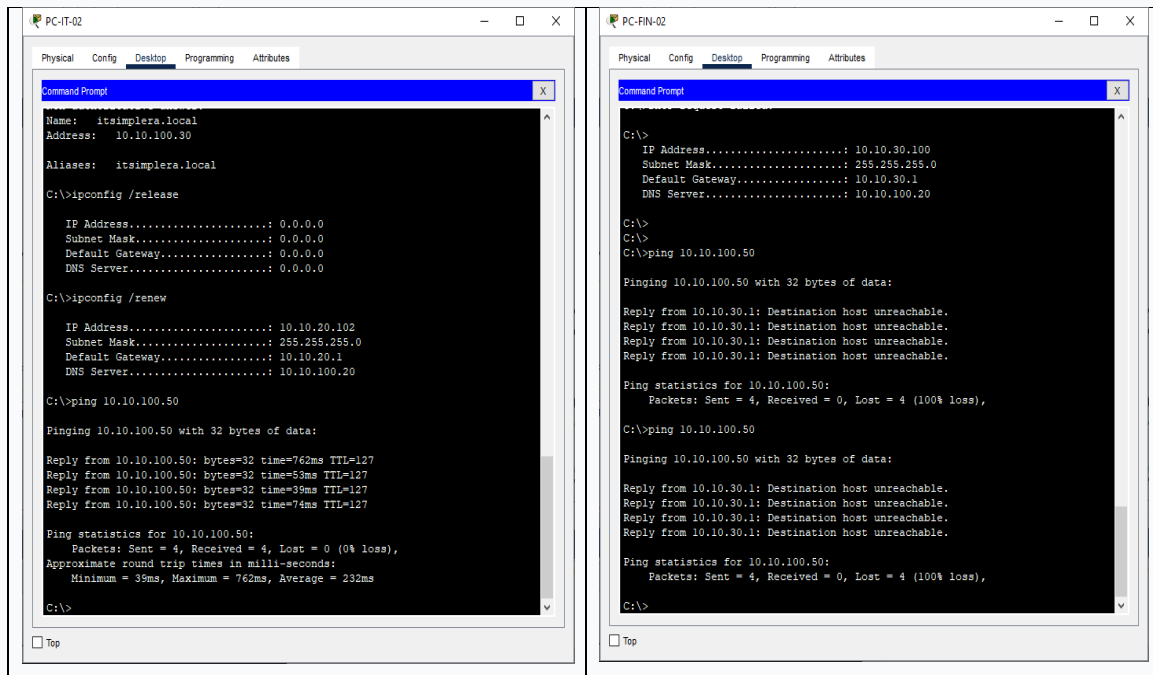
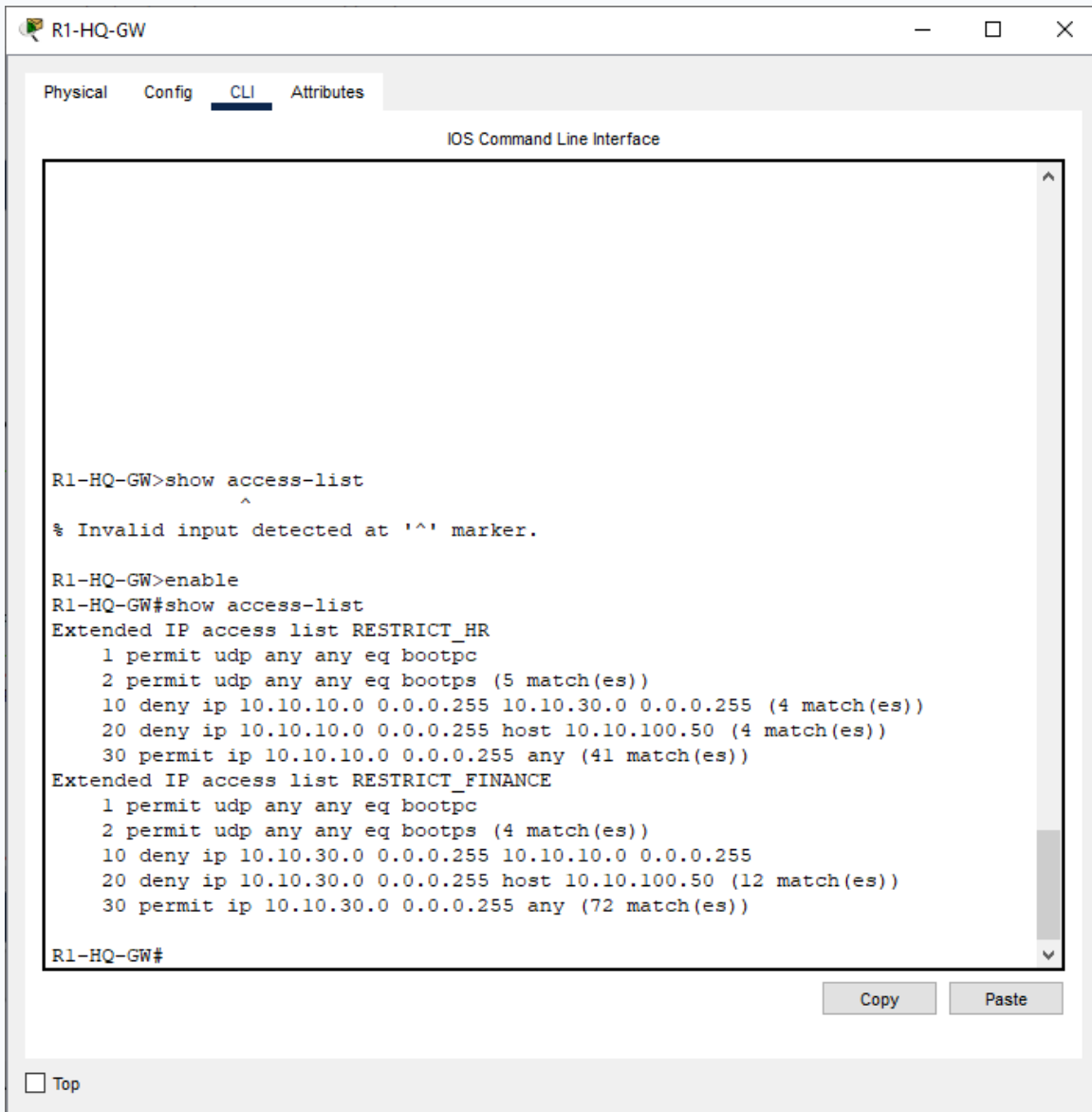


Figure 18,19: demonstrate successful administrative access from PC-IT alongside active ACL blocking (RESTRICT_FINANCE and RESTRICT_HR (Command Prompt output))

5.2 CLI Verification Commands Used

- show ip interface brief — confirm sub-interface status and IP addressing on R1-HQ-GW
- show access-lists — confirm ACL match/hit counters for RESTRICT_HR and RESTRICT_FINANCE
- show ntp status / show clock detail — confirm time synchronization
- show running-config | section logging — confirm Syslog host and trap level
- show running-config | section snmp — confirm SNMP community and host configuration
- ipconfig /all (client) — confirm DHCP-assigned addressing
- ping / tracer — confirm reachability and blocked paths across VLANs



```

R1-HQ-GW>show access-list
^
% Invalid input detected at '^' marker.

R1-HQ-GW>enable
R1-HQ-GW#show access-list
Extended IP access list RESTRICT_HR
 1 permit udp any any eq bootpc
 2 permit udp any any eq bootps (5 match(es))
10 deny ip 10.10.10.0 0.0.0.255 10.10.30.0 0.0.0.255 (4 match(es))
20 deny ip 10.10.10.0 0.0.0.255 host 10.10.100.50 (4 match(es))
30 permit ip 10.10.10.0 0.0.0.255 any (41 match(es))
Extended IP access list RESTRICT_FINANCE
 1 permit udp any any eq bootpc
 2 permit udp any any eq bootps (4 match(es))
10 deny ip 10.10.30.0 0.0.0.255 10.10.10.0 0.0.0.255
20 deny ip 10.10.30.0 0.0.0.255 host 10.10.100.50 (12 match(es))
30 permit ip 10.10.30.0 0.0.0.255 any (72 match(es))

R1-HQ-GW#

```

Figure 20: R1-HQ-GW show access-lists output showing updated ACL hit counters after Week 5 traffic

5.3 Troubleshooting Report

The table below documents the issues encountered while bringing the services online, their root cause, and the resolution applied. Replace or extend this table with the specific issues you encountered in your own lab.

Issue Observed	Root Cause	Resolution	Verified By
HR and Finance clients did not receive a DHCP lease	No ip helper-address configured on the department sub-interfaces, so DHCP broadcasts never reached SRV-DHCP across the VLAN 100 boundary	Added ip helper-address 10.10.100.20 to Gi0/0/0.10, .20 and .30	ipconfig /all on affected clients showing a valid leased address
SNMP polling from SRV-MGMT to R1-HQ-GW timed out	Community string configured on the server did not match the router's snmp-server community string	Re-matched the community string public on both ends	Successful SNMP poll returning interface and uptime data
Syslog messages were not appearing on SRV-MGMT	Default logging trap severity excluded informational-level messages	Set logging trap informational on R1-HQ-GW	%SYS-5-CONFIG_I event visible on the Syslog panel after a config change
NTP / Syslog / SNMP unreachable from HR and Finance test hosts	Expected behaviour — RESTRICT_HR and RESTRICT_FINANCE explicitly deny traffic to host 10.10.100.50	No change required; confirmed as by-design management-plane lockdown	show access-lists deny counters incrementing on each blocked attempt

CHAPTER 6

TOOLS, TECHNOLOGIES & SKILLS GAINED

6.1 Software Tools

- Cisco Packet Tracer — primary simulation platform used to build the topology and configure/verify all eight enterprise services.
- Command Prompt (client desktops) — used for ipconfig, ping, tracert, ftp, and nslookup-style verification.

6.2 Hardware / Simulated Devices

- 1 x Router-on-a-stick gateway (R1-HQ-GW) with 802.1Q sub-interfaces
- Access-layer switches for HR, IT, Finance and the Server Farm
- 5 x Generic Servers (SRV-DHCP, SRV-DNS, SRV-WEB, SRV-FILE, SRV-MGMT)
- End-user PCs and a network printer (PRINTER-HR-01) per department

6.3 Protocols & Standards Applied

DHCP, DNS, HTTP, HTTPS, FTP, TFTP, NTP, Syslog, SNMP, IEEE 802.1Q trunking, and Extended IP Access Control Lists.

6.4 Skills & Learning Outcomes

- Deploying and integrating multiple centralized network services on a single, segmented enterprise topology.
- Configuring DHCP relay (ip helper-address) to support cross-subnet address assignment.
- Building and querying DNS zone records to support name-based service access.
- Applying and validating layered security — confirming that pre-existing ACLs correctly extend to newly deployed management services (NTP, Syslog, SNMP).
- Using centralized logging (Syslog) and monitoring (SNMP) to support ongoing operational visibility.
- Producing structured technical documentation, verification matrices, and troubleshooting reports suitable for handover to a supervising administrator.

CHAPTER 7

CONCLUSION

7.2 Conclusion

Week 5 successfully extended the segmented, ACL-hardened enterprise network built in Week 4 into a fully functional services infrastructure. DHCP, DNS, HTTP/HTTPS, FTP, TFTP, NTP, Syslog, and SNMP were deployed on the Server Farm VLAN and verified as reachable from HR, IT, and Finance where intended, while the existing management-plane lockdown correctly carried over to the newly added NTP, Syslog, and SNMP services. The result is a network that is not only isolated and access-controlled, but also centrally administered, monitored, and time-synchronized reflecting real-world enterprise operational practice.